# Coverage, Context, Convention: Restoring Vowel Length in Old English

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**Abstract**

Many editions of Old English mark long vowels with macrons, but the manuscripts do not mark length systematically, so bare text merges *god* 'God' and *gōd* 'good'. We restore the marks by choosing, for each bare word, among the length patterns that candidate sources list, and we ask what limits the result: missing candidates, too little context, or the editor's convention. We build a length-marked corpus of nine editions (161,062 tokens), to our knowledge the only one besides Tauber's *Beowulf*, and release it with Tauber's text. Missing candidates are the largest limit: with WikiPron candidates, 64.8% of a frequency prior's prose errors and 73.4% of its verse errors have no correct candidate. Context helps a group of five closed-class words, chiefly *þe/þē*, and barely changes the rest: on prose, a ByT5 scorer that sees two words on each side raises accuracy on the five from 62.66% to 75.40%, and a post hoc logistic model with word-specific collocations does about as well. On ambiguous verse tokens context adds nothing, and a post hoc analysis traces this to one unseen name. Convention can outweigh context: swapping the gold between two *Beowulf* editions moves the prior's score on the five words by 11.81 points, 1.60 times the context gain on the same tokens with Klaeber as gold. CharsiuG2P's published Old English word error rate (0.208) partly reflects a 20-step decoding limit (0.188 with 50 steps).

Code and datasets are available at:
- https://github.com/Ace3Z/coverage-context-convention
- https://huggingface.co/datasets/Ace3Z/old-english-vowel-length.

# 1 Introduction

Many editions of Old English mark long vowels with a macron. The manuscripts do not mark length systematically, though some Late Old English scribes marked it with accents [17]. Bare text merges *god* 'God' and *gōd* 'good', or the relative particle *þe* and the pronoun *þē* 'thee', so restored marks could help readers of texts transcribed without them. Restoring them means choosing, for each word, among the length patterns that candidate sources list: lexical disambiguation as in accent restoration [34] and homograph disambiguation in text-to-speech [12], here for a historical language with little data. The ambiguity is lexical (class A below, *god*), grammatical (class B, *þe*) or editorial (class C). We predict an editor's marking convention, not pronunciation.

To our knowledge, no system has restored Old English vowel length in context. We measure three limits on the same two test texts. *Coverage* (Section 5.1) asks whether the right candidate exists, and with WikiPron candidates, for most errors of a frequency prior (which picks each word's most frequent pattern), it does not. *Context* (Section 5.2) asks whether neighbouring words can pick it out, and they help mainly class B, chiefly *þe/þē*. *Convention* (Section 5.3) asks whose marking the gold follows, and on class B the choice of *Beowulf* edition moves the prior more than context adds. We contribute a length-marked corpus of nine editions (Section 2, to our knowledge the only one besides Tauber's *Beowulf* [25]), an evaluation that separates the three limits, a reanalysis of CharsiuG2P [36] (Section 5.4) and tested code whose one script recomputes every result here from the released data and model predictions (MIT licence).

**Related work.** Yarowsky [34] restored Spanish and French accents with decision lists over collocations, taking accented text as a fully objective standard, which Old English editions are not. Latin [30] and Ancient Greek [27] macronisers use taggers, lemmatisers and morphological analysers or large pretraining [7]. Winge [30] already bounded his Latin system with a lexicon oracle and traced the largest group of its disagreements with a hand-marked text to marking conventions. Gershuni and Pinter [11] restore Hebrew diacritics with a character tagger instead of candidates, and Esmail et al. [10] measure how much lookahead helps Arabic diacritisation. We test lookahead with a model that sees only preceding words. Plank [21] argues that one gold label hides annotator variation, which here includes editorial convention.

# 2 Data set

Table 1 lists the corpus: nine editions printed from 1880 to 1921 with 161,062 tokens in 7,889 sentences, plus Tauber's *Beowulf*. We took Project Gutenberg and Wikisource transcriptions [2, 3, 4, 6, 13, 14, 24, 32, 37] and cleaned them with character mappings, each counted in a manifest (some files cannot be rebuilt exactly from this repository). The Hugging Face release (CC BY-SA 4.0) keeps each sentence's work, editor, year, genre and section, per-token length patterns and a leakage flag. We built the corpus because word lists lack context, treebanks such as YCOE [26] carry, as far as we know, no length marks, and Tauber's *Beowulf* is a single poem. Our corpus has 9.3 times its tokens, and together they hold three editions of *Beowulf* to compare.

Splits are by work, so no test work is seen in training. Training (69,087 tokens, without Zupitza, Section 4) is 90.9% prose and 85.6% Bright's. The development text, John, shares a Gospel series with the Matthew training text and is easier than the tests: 6.36% of its tokens are unseen in training, against 17.94% (prose test) and 31.53% (verse test). A word 6-gram audit removes 34 John sentences and 1 *Soliloquies* sentence from scoring, and the longest word run that *Beowulf* shares with training has 5 words. The verse gold is Brunetti's tokenisation of *Beowulf* with the lengths of the Perseus text of Klaeber's edition [15], transferred by Tauber.

# 3 Preprocessing

Preprocessing normalises Unicode (NFC), keeps the macron as the only diacritic, drops English and Latin lines by their function words, splits sentences at end punctuation and takes letter runs as tokens. A token's *key* is its lower-cased form without macrons, and its *pattern* has one digit per vowel letter, 1 for a macron: Harrison and Sharp's *geārdagum* (0100) and Klaeber's *gēardagum*

**Table 1:** The corpus. PG: Project Gutenberg, WS: Wikisource. % long: share of vowel letters marked long.

| Edition | Editor, printing | Source | Genre | Tokens | % long | Role (% unseen) |
|---|---|---|---|---|---|---|
| Ælfric's *Grammar* | Zupitza 1880 | PG | prose | 23,940 | 23.58 | ablation |
| *Anglo-Saxon Reader* | Bright 1913 | WS | mixed | 38,484 | 23.91 | train |
| Gospel of Matthew | Bright, 1910 printing | WS | prose | 20,648 | 25.86 | train |
| *Anglo-Saxon Primer* | Sweet 1905 | PG | prose | 9,003 | 24.20 | train |
| *Physiologus* | Cook 1921 | PG | verse | 952 | 20.57 | train |
| Gospel of John | Bright 1904 | WS | prose | 17,310 | 26.14 | dev (6.36) |
| Alfred's *Soliloquies* | Hargrove 1902 | WS | prose | 15,475 | 26.11 | prose test (17.94) |
| *Beowulf* | Harrison & Sharp 1894[a] | PG | verse | 17,682 | 21.24 | agreement |
| *Beowulf* | Wyatt 1894[b] | WS | verse | 17,568 | 20.74 | agreement |
| *Beowulf* | Klaeber 1922 | Tauber | verse | 17,361 | 21.22 | verse test (31.53) |

[a]Project Gutenberg revision of the 4th edition (first edition 1883). [b]Its *Finnsburh Fragment* lacks lines 1 to 18 in our file (187 of 297 tokens). Scored tokens: dev 16,649, prose 15,426, verse 17,361. Training without Zupitza: 69,087 tokens. Hugging Face release: 8,650 sentences, 178,423 tokens.

(1000) share the key *geardagum* and differ only in which digraph letter bears the macron.

A key's candidates are its training patterns plus those of every listed form with that key. Our main list is WikiPron [16], whose IPA we convert to spellings by aligning vowel units with IPA nuclei (97.6% of 55,835 rows align). A coverage comparison adds the dictionaries of our proposal, Clark Hall [22], Wiktionary via kaikki.org [35] and Bosworth-Toller [28], though WikiPron is itself mined from Wiktionary (99.68% of its forms are in the Wiktionary list). A token is *lexicon-ambiguous* when its key has two or more candidates, as for 37.8% of prose and 25.4% of verse tokens with WikiPron. As adding sources enlarges this set, we compare sources on all tokens. The 60 most frequent ambiguous keys (ranked before we dropped Zupitza, with test tokens counted but not their gold) carry a class: 13 lexical (A: *god/gōd*), 5 grammatical (B: *þe, ne, ge, gyt, þis*) and 42 editorial (C: *se/sē, he/hē*), covering 85.0% of prose and 78.4% of verse ambiguous tokens.

**Labels.** The classes, the types of 400 sampled errors and a sample of 100 editor disagreements were labelled by a large language model, Claude Opus 5.5 [1], run as separate annotator agents with written guidelines. For the classes and the error types, two runs of the model labelled every item separately (Cohen's $\kappa$ 0.927 and 0.975 [5]), and a third run settled their disagreements. A single run labelled the disagreement sample. A blind review by further runs of the same model changed 7 of the 560 labels. The labels were then manually checked, and none was changed. As both annotators are runs of one model, $\kappa$ measures that model's consistency rather than agreement between people, and as the manual check was not blind to the model's labels, we lack the comparison with independent human labels that Pangakis et al. [18] call for.

## 4 Model training

**Baselines.** All our systems are supervised, choose among the same candidates and leave a word without candidates unmarked. The frequency prior, fixed in our proposal after Yarowsky [34], picks a word's most frequent training pattern and, for an unseen word, the pattern most sources list, with ties going to fewer long vowels. Logistic classifiers (scikit-learn [20], untuned, three seeds) see sparse indicators of the word and its first and last two letters (*word*), plus the two preceding words (*left*), plus the two following words (*bilateral*). These context features are not conjoined with the word, so 'previous word *tō*' shifts every word's scores alike. After the ByT5 results we added a post hoc *conjoined* variant with the word-by-neighbour features of earlier work [12, 34].

**ByT5.** As our main model we fine-tune ByT5-small [33] with PyTorch [19] and Transformers [31] to score candidates. ByT5 reads bytes, so thorn, eth and ash need no vocabulary. It is also CharsiuG2P's backbone, and scoring candidates guarantees a valid output. The input is the bare word, alone or in context with the target marked `[[ ]]`, the output is its pattern string, and a candidate's score is the summed log-probability of its bytes, free of length bias since candidates are equally long. Arms differ only in the input: *word*, *left* (all preceding words), *sentence*, *local2* and *local5* (two or five words on each side), all trained on the same 25,949 ambiguous training tokens as the logistic models. We report the full input cap (2,560 bytes), as 128- and 256-byte

caps move mean ambiguous-token accuracy by under two points.

**Protocol.** We chose the setting (learning rate $3 \cdot 10^{-4}$, 5 epochs, no Zupitza) from 24 sentence-arm runs on development data and reused it for every arm. Dropping Zupitza's *Grammar*, which marks *þes*, *gyt* and *lytel* short, lifts the prior on ambiguous development tokens from 95.12% to 95.60% and the bilateral model from 94.70% to 95.31%. ByT5 also gains slightly in all but the least trained of the 5 settings run both ways: the change spans $-7.40$ to $+0.49$ points (we leave out a sixth, where one run chose the candidate with most long vowels for 98.21% of ambiguous development tokens). The price, measured afterwards, is coverage: only Zupitza writes *forþām*, *swāswā* and *forþī* as one word, so 153 prose test tokens become unknown and the prior loses 0.88 points. The grid has 39 runs (four context arms at three caps and the word arm, three seeds each). About fifty exploratory runs were scored on the test sets before the development set existed, and the selection runs also logged test scores. The choice used development scores only.

**Changes from the proposal.** The corpus grew to nine editions. A ByT5 encoder-decoder replaced the planned encoder classifier. WikiPron superseded the proposed dictionaries as the main candidate source. We added the logistic baselines and the local2 and local5 arms, and chose local2 as the headline arm after the test results. The class B focus and the gold swap of Section 5.3 were not planned. An LLM labelled what we had planned to label ourselves. We set the threshold for reporting editor disagreement instead of a model only after measuring, so it decided nothing.

# 5 Evaluation

Our main finding: missing candidates cost more than context adds on both test texts, and on the *Beowulf* class B tokens the choice of gold edition moves the prior's score more than context adds. We score token accuracy, as a reader needs the whole word: a token counts only if every vowel has the gold length. On all tokens, marking nothing scores 57.58% (prose) and 59.11% (verse), the WikiPron prior 87.85% and 89.40%, and local2 88.91% and 89.75%. Differences carry 95% intervals from a paired bootstrap over sentences [9], whose tokens are not independent (1,000 draws, correctness averaged over runs, runs held fixed). Intervals and $p$-values are not corrected for multiple comparisons, and we claim no difference between ByT5 arms below about one point.

## 5.1 Coverage

Most of the prior's errors are beyond any chooser: with WikiPron, 64.8% of its prose errors and 73.4% of its verse errors have no correct candidate, mostly because the word is unknown. Our proposal had assumed that every error is a selection error. Both tests have about the same achievable ceiling, a perfect chooser's score (Figure 1, 92.13% and 92.22%), and a better chooser could gain at most 4.28 points over the prior on prose and 2.82 on verse. Dictionaries change the prior only for words unseen in training, as a dictionary-only pattern has no training count: on unseen verse tokens the prior rises from 52.89% to 85.02%. Of the verse gain from all four sources (10.13 points), 94% falls on *Beowulf*-only word types, so it may not hold for other verse.

## 5.2 Context

Context helps class B and barely moves classes A and C. On prose, local2 (chosen after the test results) beats the word arm by $+2.39$ $[+1.96, +2.88]$ points on ambiguous tokens (Table 2), almost all in class B. There it lifts accuracy from 62.66%, where the word arm equals the prior, to 75.40% ($+12.74$ $[+10.37, +15.03]$, or $[+8.84, +16.84]$ with runs resampled). Every context arm shows this gain, even the left arm, which sees no following word. Most of it comes from telling relative *þe* from the pronoun *þē* 'thee' (57.7% to 76.5% on 567 tokens). In *Ic clypie tō þē, Drihten, þū þe æall geworhtest* 'I call to thee, Lord, thou who madest all', the prior, the ByT5 word runs and the logistic word, left and bilateral runs mark both *þe* short, while every local2 run and every conjoined run gets the pronoun long and the relative short.

On verse, one name masks the context gain: local2 gains $+7.22$ $[+4.74, +9.83]$ points on class B, mostly from *þe*, yet on ambiguous verse tokens no context arm clearly beats the word arm (local2 $+0.29$ $[-0.11, +0.64]$). *Gēata* 'of the Geats' (39 tokens, all long) is unseen in training, and

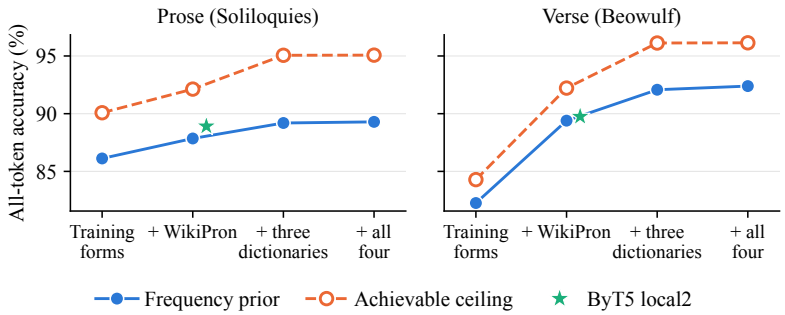



**Figure 1:** Coverage. All-token accuracy of the frequency prior and the achievable ceiling (gold among the candidates, or no candidate and an all-short gold). '+ three dictionaries' leaves out WikiPron (derived from Wiktionary), and '+ all four' adds it. Star: ByT5 local2 with WikiPron candidates, mean of three runs.

**Table 2:** Context. Accuracy (%) at the full byte cap, mean over three runs, ± SD for lexicon-ambiguous (Lex.-amb.) tokens and class B. Classes A, B, C: LLM labels, manually checked. †Post hoc diagnostic.

| | Prose (*Soliloquies*) | | | | Verse (*Beowulf*) | | | |
|---|---|---|---|---|---|---|---|---|
| | Lex.-amb. | A | B | C | Lex.-amb. | A | B | C |
| *n* | 5,834 | 663 | 1,023 | 3,275 | 4,402 | 387 | 420 | 2,645 |
| Frequency prior | 88.46 | 97.44 | 62.66 | 95.85 | 88.85 | 89.41 | 71.19 | 95.80 |
| Logistic bilateral | 89.67±0.05 | 96.88 | 67.64±0.59 | 95.85 | 88.69±0.08 | 89.41 | 73.33±0.24 | 95.80 |
| Logistic conjoined† | 91.26±0.15 | 97.44 | 76.02±0.96 | 95.85 | 89.15±0.11 | 89.41 | 77.38±0.95 | 95.80 |
| ByT5 word | 88.88±0.08 | 97.44 | 62.66±0.00 | 95.85 | 89.97±0.03 | 89.41 | 71.19±0.00 | 95.80 |
| ByT5 left | 91.33±0.36 | 97.34 | 76.87±1.98 | 95.85 | 89.54±0.06 | 89.49 | 76.51±0.96 | 95.83 |
| ByT5 sentence | 90.34±1.26 | 97.49 | 70.90±7.09 | 95.85 | 89.29±0.13 | 89.41 | 75.08±3.42 | 95.90 |
| ByT5 local2 | 91.27±0.52 | 97.13 | 75.40±3.32 | 95.88 | 90.26±0.44 | 89.41 | 78.41±2.38 | 95.80 |
| ByT5 local5 | 91.85±0.76 | 97.44 | 79.05±3.87 | 95.86 | 89.75±0.16 | 89.41 | 77.62±2.65 | 95.80 |

WikiPron also lists *geata* 'gates'. The prior falls back to fewer long vowels and misses every token, the word arm gets all in every seed, and the sentence arm almost none. Without *Gēata* (post hoc), local2 minus word is +0.78 [+0.43, +1.12] and sentence minus word +0.18 [−0.16, +0.48], both below our one-point margin. On all prose tokens local2 adds 1.06 points to the WikiPron prior (sentence arm 0.71) out of the 4.28 a better chooser could add. The three other dictionaries add more (1.44). On all verse tokens the word arm alone does as well as local2 (89.68% and 89.75%).

The bilateral logistic model gains less over its word model (prose class B +4.99 [+3.58, +6.36]), as its features cannot express that 'previous word *tō*' matters for *þe* alone. The post hoc conjoined variant can, and on prose we cannot tell it from local2 (local2 minus conjoined on class B: −0.62 [−2.31, +1.05]), while on ambiguous verse tokens local2 leads by about a point (+1.11 [+0.65, +1.55]). Training variance limits the table: over the sentence arm's nine runs at the three caps, prose class B spans 62.85% to 81.43%. Local inputs never reach a cap, so local runs of one seed repeat the same training at each cap, yet local5 replicates differ by up to 6.84 points on prose class B. Seeds [8] and tiny nondeterminism [23] move training results. We set no deterministic mode and did not test the cause, and three runs per arm cannot rank the context arms.

## 5.3 Convention

On class B, which edition serves as gold matters more than context. Table 3 compares the three *Beowulf* editions on the tokens each pair aligns (identical spelling runs, 93.8% to 97.4% of each text). Our Harrison and Sharp file, a Project Gutenberg revision that modernised some quantities (conjunctive *ne* to *nē*), sits close to Klaeber, so Wyatt against Klaeber is our two-editor reference. They agree on 96.08% of ambiguous tokens but only on 78.07% of class B. Both texts nearly always mark the pronouns *þē* 'thee' and *gē* 'ye' long, but in 89 of their 91 class B disagreements Klaeber marks a *nē*, *gȳt* 'yet', *gē* 'and' or relative *þē* long and Wyatt's text does not. Of a labelled sample of 100 disagreements between Wyatt's text and Harrison and Sharp's, 54 are editorial, 10 put the macron on the other letter of a digraph and 36 are other, mostly apparent slips in one text, so the editions probably agree on length more often than we measure. On the 415 shared class B tokens, the prior agrees with Wyatt on 82.89% and with Klaeber on 71.08%, a swing of 11.81 points.

**Table 3:** Agreement (%, $\kappa$ of whole-token patterns in parentheses) on aligned *Beowulf* tokens (H&S, Wyatt: also 177 *Finnsburh* tokens). Lex.-amb.: lexicon-ambiguous. H&S: Harrison and Sharp.

| | H&S, Wyatt | H&S, Klaeber | Wyatt, Klaeber |
|---|---|---|---|
| Aligned | 16,833 | 16,577 | 16,918 |
| All | 97.73 (0.973) | 98.82 (0.986) | 97.67 (0.972) |
| Lex.-amb. | 96.08 (0.942) | 98.82 (0.983) | 96.08 (0.942) |
| *n* | 4,387 | 4,334 | 4,335 |
| Class B | 78.52 (0.469) | 98.79 (0.974) | 78.07 (0.460) |
| *n* | 419 | 412 | 415 |

**Table 4:** Sampled error types (LLM labels, manually checked, $\kappa$ 0.986 for ByT5, 0.963 for CharsiuG2P).

| | ByT5 | | CharsiuG2P | |
|---|---|---|---|---|
| Type | Prose | Verse | Prose | Verse |
| Gold not a candidate | 3 | 0 | 0 | 0 |
| Another edition agrees | 0 | 12 | 0 | 5 |
| Homograph (A) | 9 | 48 | 26 | 29 |
| Grammatical (B) | 59 | 14 | 50 | 23 |
| Editorial (C) | 33 | 22 | 35 | 32 |
| Other | 0 | 0 | 0 | 0 |
| Items | 104 | 96 | 111 | 89 |

Against Wyatt the prior even beats the editors' own agreement, which is therefore no ceiling on accuracy. Local2 gains +7.39 [+4.87, +10.25] points over the word arm with Klaeber as gold and +2.41 [−0.40, +5.42] with Wyatt. The prior's swing is thus 1.60 times the larger context gain, and local2's own score moves with the gold too (85.30% against Wyatt, 78.47% against Klaeber).

The article *se* shows convention most clearly. Klaeber writes it short in 102 of 105 article tokens and long in all 70 pronoun and relative tokens (by Brunetti's glosses), while our training editions write *sē* 823 times against 115 *se*. The prior gets 0 of the short articles right and no ByT5 run more than 14, and in line 2587, *þæt se mǣra maga Ecgðēowes*, the prior and all 39 ByT5 runs mark it long. Klaeber's choice follows syntax and could be learned from Klaeber-style text, but a model cannot tell from bare text which convention the gold follows.

### 5.4 Errors, CharsiuG2P and limitations

ByT5's prose errors are most often grammatical (59 of 104, *þe* 34 times) and its verse errors most often homographs (48 of 96, *Gēata* 10 times). Table 4 classifies 200 random ambiguous errors of the ByT5 sentence arm (seed 1, full cap) and 200 of CharsiuG2P (with Zupitza in the lexicon). The two systems' profiles over A, B and C differ within verse ($\chi^2$ test in SciPy [29], $p = 0.013$) and within prose ($p = 0.014$), in opposite directions, but not when pooled ($p = 0.62$). A fixed rule decides 279 of the 400 items, so label agreement is high partly by construction ($\kappa$ 0.945 on the 130 items without a class). In line 2796, *ēcum dryhtne þē ic hēr on starie*, every full-cap ByT5 run writes Wyatt's *þe* and is scored wrong against Klaeber's *þē*.

**CharsiuG2P.** CharsiuG2P [36] is a multilingual ByT5 model that reads isolated words, trained on pronunciation lists (WikiPron's for Old English). It was built for dictionary pronunciation, so our scores measure how well it fits our editors' marking in running text. We run it as published (prefix `<ang>:`) and map its IPA to patterns with our aligner, counting unmappable outputs as wrong. We reproduce its published Old English scores exactly (WER 0.208, PER 0.0410). Its evaluation stops decoding at Transformers' default of 20 steps, truncating 12 of 500 outputs. With 50 steps the WER is 0.188, and 71 of 94 word errors are length-only. On our text it gets length right on 70.96% of prose and 77.78% of verse tokens, well below the prior, and of CharsiuG2P's 1,580 ambiguous prose length errors the context-free prior repairs 87.09% from word habit alone.

**Limitations.** One test text per genre lets single words decide results (*Gēata*, *þe*), and 85.6% of training is Bright's, as is the development text, so choices made on it may favour his conventions. The gold follows one editor, and we have not proofread the transcriptions against the print. An LLM made the labels, and the manual check was not blind and had no second annotator, so no agreement between human annotators was measured.

## 6 Future work

If convention explains most *Beowulf* class B and *se* errors, training and testing in one editorial tradition should reduce them, and scoring against several editions would show what remains. The main caveats call for proofread test texts, ByT5 with all four sources, a dictionary-free tagger [11] for unlisted patterns (weaker than the best dictionary-based systems on unseen words), deterministic training [23] with five or more seeds, more verse and independent human labels for a sample.

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## Use of AI tools

We used an AI assistant (Claude, Anthropic) for help with editing the code and checking the diction of this report, and to produce the labels described in Sections 3 and 5, which were then manually checked. We directed the work and take responsibility for its content.